```
ubuntu@ubuntu:~/mininet$ sudo mn -c
*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes
killall controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd ovs-contro
llerovs-testcontroller udpbwtest mnexec ivs ryu-manager 2> /dev/null
killall -9 controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd ovs-con
trollerovs-testcontroller udpbwtest mnexec ivs ryu-manager 2> /dev/null
pkill -9 -f "sudo mnexec"
*** Removing junk from /tmp
rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log
*** Removing old X11 tunnels
*** Removing excess kernel datapaths
ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'
***  Removing OVS datapaths
ovs-vsctl --timeout=1 list-br
ovs-vsctl --timeout=1 list-br
*** Removing all links of the pattern foo-ethX
ip link show | egrep -o '([-_.[:alnum:]]+-eth[[:digit:]]+)'
ip link show
*** Killing stale mininet node processes
pkill -9 -f mininet:
*** Shutting down stale tunnels
pkill -9 -f Tunnel=Ethernet
pkill -9 -f .ssh/mn
rm -f ~/.ssh/mn/*
*** Cleanup complete.
```

```
ubuntu@ubuntu:~/mininet$ sudo mn --topo single,3 --controller=remote
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
Connecting to remote controller at 127.0.0.1:6633
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> sh ovs-ofctl show s1
OFPT_FEATURES_REPLY (xid=0x2): dpid:0000000000000001
n_tables:254, n_buffers:0
capabilities: FLOW_STATS TABLE_STATS PORT_STATS QUEUE_STATS ARP_MATCH_IP
```

```
capabilities: FLOW_STATS TABLE_STATS PORT_STATS QUEUE_STATS ARP_MATCH_IP
actions: output enqueue set_vlan_vid set_vlan_pcp strip_vlan mod_dl_src mod_dl_dst mod_nw_
src mod_nw_dst mod_nw_tos mod_tp_src mod_tp_dst
 1(s1-eth1): addr:96:37:52:c1:d2:57
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 2(s1-eth2): addr:da:91:83:f3:1f:9f
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 3(s1-eth3): addr:ee:2d:af:ca:45:bc
     config:     0
     state:      0
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 LOCAL(s1): addr:06:ac:60:23:77:40
     config:     PORT_DOWN
     state:      LINK_DOWN
     speed: 0 Mbps now, 0 Mbps max
OFPT_GET_CONFIG_REPLY (xid=0x4): frags=normal miss_send_len=0
mininet> sh ovs-ofctl add-flow s1 priority=100,in_port=1,actions=drop
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> sh ovs-ofctl del-flows s1
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> sh ovs-ofctl del-flows s1
mininet> sh ovs-ofctl add-flow s1 priority=100,ip,nw_src=10.0.0.1,nw_dst=10.0.0.2,actions=
drop
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.

^C
--- 10.0.0.2 ping statistics ---
48 packets transmitted, 0 received, 100% packet loss, time 47202ms
```

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> ^C
Interrupt
stopping h1
mininet> h1 ping h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=6.63 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=1.26 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.086 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.209 ms
64 bytes from 10.0.0.3: icmp_seq=5 ttl=64 time=0.211 ms
64 bytes from 10.0.0.3: icmp_seq=6 ttl=64 time=0.063 ms
64 bytes from 10.0.0.3: icmp_seq=7 ttl=64 time=0.179 ms
^C
--- 10.0.0.3 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6009ms
rtt min/avg/max/mdev = 0.063/1.233/6.625/2.234 ms
mininet> pingall
*** Ping: testing ping reachability
h1 -> X h3
h2 -> X h3
h3 -> h1 h2
*** Results: 33% dropped (4/6 received)
```

```
mininet> sh ovs-ofctl dump-flows s1
 cookie=0x0, duration=135.809s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_s
rc=46:fe:6f:96:5c:d2,dl_dst=4a:fa:c6:5d:78:23,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,arp_op=1 a
ctions=output:"s1-eth2"
 cookie=0x0, duration=135.761s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_s
rc=4a:fa:c6:5d:78:23,dl_dst=46:fe:6f:96:5c:d2,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,arp_op=2 a
ctions=output:"s1-eth1"
 cookie=0x0, duration=67.725s, table=0, n_packets=2, n_bytes=84, arp,vlan_tci=0x0000,dl_sr
c=2a:ca:e5:67:d7:65,dl_dst=46:fe:6f:96:5c:d2,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,arp_op=1 ac
tions=output:"s1-eth1"
 cookie=0x0, duration=67.725s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_sr
c=46:fe:6f:96:5c:d2,dl_dst=2a:ca:e5:67:d7:65,arp_spa=10.0.0.1,arp_tpa=10.0.0.3,arp_op=1 ac
tions=output:"s1-eth3"
 cookie=0x0, duration=67.684s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_sr
c=46:fe:6f:96:5c:d2,dl_dst=2a:ca:e5:67:d7:65,arp_spa=10.0.0.1,arp_tpa=10.0.0.3,arp_op=2 ac
tions=output:"s1-eth3"
 cookie=0x0, duration=67.684s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src
=2a:ca:e5:67:d7:65,dl_dst=46:fe:6f:96:5c:d2,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,arp_op=2 act
ions=output:"s1-eth1"
 cookie=0x0, duration=48.245s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src
=4a:fa:c6:5d:78:23,dl_dst=46:fe:6f:96:5c:d2,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,arp_op=1 act
ions=output:"s1-eth1"
 cookie=0x0, duration=48.198s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src
=46:fe:6f:96:5c:d2,dl_dst=4a:fa:c6:5d:78:23,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,arp_op=2 act
ions=output:"s1-eth2"
 cookie=0x0, duration=38.538s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_sr
c=2a:ca:e5:67:d7:65,dl_dst=4a:fa:c6:5d:78:23,arp_spa=10.0.0.3,arp_tpa=10.0.0.2,arp_op=1 ac
tions=output:"s1-eth2"
```

```
mininet> exit
*** Stopping 1 controllers
c0
*** Stopping 3 links
...
*** Stopping 1 switches
s1
*** Stopping 3 hosts
h1 h2 h3
*** Done
completed in 320.169 seconds
```

```
ubuntu@ubuntu:~/mininet$ sudo mn --topo single,3 --controller=remote
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
Connecting to remote controller at 127.0.0.1:6633
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> sudo mn --topo single,3 --controller=remote
*** Unknown command: sudo mn --topo single,3 --controller=remote
mininet> sh ovs-ofctl dump-flows s1
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=46.5 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.498 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.330 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.251 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.321 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.086 ms
```

```
mininet> sh ovs-ofctl dump-flows s1
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=46.5 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.498 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.330 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.251 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.321 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.086 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.332 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.137 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.263 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.248 ms
iperf64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.189 ms
^?^?^?^?64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.097 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.058 ms
^C
--- 10.0.0.2 ping statistics ---
13 packets transmitted, 13 received, 0% packet loss, time 12032ms
rtt min/avg/max/mdev = 0.058/3.795/46.525/12.335 ms
mininet> iperf
*** Iperf: testing TCP bandwidth between h1 and h3
^C
Interrupt
stopping h3
mininet> sh ovs-ofctl del-flows s1
mininet>
```